

Introduction

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Brief Overview of the Course

- Economics suggests important relationships, often with policy implications, but virtually never suggests quantitative magnitudes of causal effects.
- What is the quantitative effect of reducing class size on student achievement?
- How does another year of education change earnings?
- What is the effect on output growth of a 1 percentage point increase in interest rates by the Fed?
- What is the effect on housing prices of environmental improvements?
- What is the effect of adding more bidders to an auction on the selling price?

Brief Overview of the Course

Suppose you want to answer the following question: *What is the quantitative effect of reducing class size on student achievement?*

- Ideally, we would like to design and experiment.
- What would be an experiment to estimate the effect of class size on standardized test scores?

But almost always we only have observational (nonexperimental) data.

- Returns to education.
- Cigarette prices.
- Monetary policy.

Brief Overview of the Course

Most of the course deals with difficulties arising from using observational data to estimate causal effects.

- confounding effects (omitted factors).
- simultaneous causality (nonexperimental) data.
- “correlation does not imply causation”.

Brief Overview of the Course

In this course you will:

- Learn to evaluate the regression analysis of others; this means you will be able to understand empirical economics papers you will read in other Econ courses (or at work).
- Get some hands-on experience with regression analysis.
- Learn methods for estimating causal effects using observational data.
- Learn some tools that can be used for other purposes; for example: experiments.
- Focus on applications: theory is used only as needed to understand the whys of the methods.

Brief Overview of the Course

Example: Class Size and Educational Output

- Policy question: What is the effect on test scores (or some other outcome measure) of reducing class size by one student per class?
- We must use data to find out.

The California Test Score Data Set

Class size and educational output

All K-6 and K-8 California school districts ($n = 420$)

- 5th grade test scores (Stanford-9 achievement test, combined math and reading) district average.
- Student-teacher ratio (STR) = number of students in the district divided by the number of full-time equivalent teachers.

First Look at the Data

TABLE 4.1 Summary of the Distribution of Student–Teacher Ratios and Fifth-Grade Test Scores for 420 K–8 Districts in California in 1999

	Average	Standard Deviation	Percentile						
			10%	25%	40%	50% (median)	60%	75%	90%
Student–teacher ratio	19.6	1.9	17.3	18.6	19.3	19.7	20.1	20.9	21.9
Test score	654.2	19.1	630.4	640.0	649.1	654.5	659.4	666.7	679.1

- Do you remember percentiles?
- For example, 50% of the districts have student-teacher ratios below 19.7 students-per-teacher.

First Look at the Data

What does this figure show?

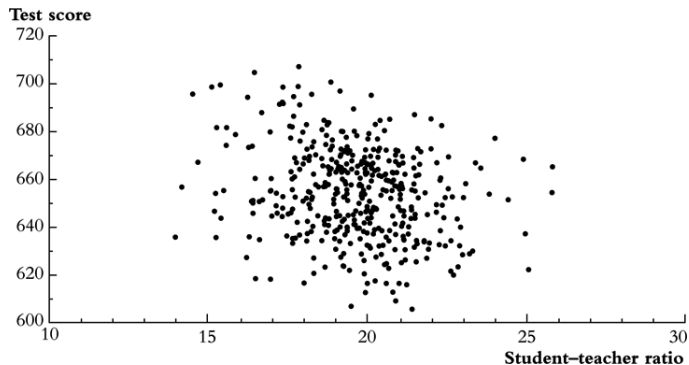


Figure : Test Score vs. Student-Teacher Ratio

The California Test Score Data Set

How to Proceed

We need to get some numerical evidence on whether districts with low STRs have higher test scores—but how?

- 1 Compare average test scores in districts with low STRs to those with high STRs (**estimation**).
- 2 Test the “null” hypothesis that the mean test scores in the two types of districts are the same, against the “alternative” hypothesis that they differ (**hypothesis testing**).
- 3 Estimate an interval for the difference in the mean test scores of *high* vs. *low* STR districts (**confidence interval**).

The California Test Score Data Set

Initial Data Analysis

- Define two groups depending on the student-teacher-ratio (STR).
- For example: call “small” to those with $STR < 20$ and “large” to those with $STR \geq 20$.

Class size	Average score	Standard deviation	Sample size
Small	657.4	19.42	38
Large	650	17.9	182

- 1 Estimate the difference between group means Δ .
- 2 Test the hypothesis: $\Delta = 0$.
- 3 Construct a confidence interval for Δ .

Econometrics

- Econometrics is the science and art of using economic theory and statistical techniques to analyze economic data.
- Econometric methods are used in many branches of economics and business as well as in other social sciences, including political science and sociology.

Sample Questions

In this chapter we will discuss four questions:

- Does reducing class size improve elementary school education?
- Is there racial discrimination in the market for home loans?
- How much do cigarette taxes reduce smoking?
- What will the rate of inflation be next year?

Question #1: Does reducing class size improve elementary school education?

- Argument in favor: with fewer students in the classroom each student gets more of the teacher's attention, there are fewer class disruptions, learning is enhanced, and grades improve.
- Arguments against: more teachers are needed, more classrooms may need to be built, and therefore costs increase.
- To weight costs and benefits, decision makers need a quantitative understanding of the benefits of a reduced size classroom.

Question #1: Does reducing class size improve elementary school education?

- Data collected for 420 school districts in California show that students in small classroom tend to perform better on standardized tests than students in districts in larger classrooms.
- However, this observation may be reflecting many other advantages that students in districts with smaller classrooms have.
- For example, districts with smaller classrooms tend to have wealthier residents than districts with larger classrooms.

Question #2: Is there racial discrimination in the market for home loans?

- Most of house buyers in the U.S. buy their homes with the help of a loan or a mortgage.
- By law, U.S. lending institutions cannot discriminate on the basis of race.
- However, data reveals that 28% of black applicants are denied mortgages while only 9% of white applicants are denied.

Question #2: Is there racial discrimination in the market for home loans?

- Do these data indicate that there is racial discrimination in mortgage lending?
- The fact that more black than white applicants are denied does not by itself provide evidence of discrimination because black and white applicants differ in many other ways than race.

Question #3: How much do cigarette taxes reduce smoking?

- Cigarette smoking is characterized by negative externalities. One of the ways governments have been dealing with this issue is by increasing taxes on cigarette consumption.
- How effective this policy is depends on the price elasticity of demand for cigarettes.

Question #3: How much do cigarette taxes reduce smoking?

- Data on cigarette prices, taxes, and income for U.S. states in the 1980s and 1990s shows that states with low taxes, and thus low cigarette prices, have high smoking rates, and states with high prices have low smoking rates.
- In this data, causality runs both ways: low taxes lead to high demand, but if there are many smokers in the state politicians may try to keep taxes low.

Question #4: What will the rate of inflation be next year?

- Inflation imposes cost on society. Predicting future inflation is on the interest of society as well as on the individuals.
- Econometrics can be used to forecast the future value of certain variables using historical and current data.

- Each of these questions require a numerical answer. Economic theory provides clues about that answer, but the actual value of the number should be learned empirically (i.e., using data).
- Since different data sets produce different numbers we are also interested in a measure of precision for the numerical answer.
- The conceptual framework that will give us both a numerical answer and an assessment of its precision is the multiple regression model.
- The multiple regression model provides a mathematical way to quantify how a change in one variable affects another variable, holding other things constant (or as econometricians would say: “controlling”).

Causal Effects and Idealized Experiments

Causality

- An action is said to cause an outcome if the outcome is the direct result, or consequence, of that action.
 - Eating bad food causes you to get sick.
 - Driving recklessly causes a car crash.
 - Reading a good book causes pleasure.

Causal Effects and Idealized Experiments

Estimation of Causal Effects: Example

- Measuring the causal effect on tomato yield (measured in grams) of applying a certain amount of fertilizer, say 100 grams of fertilizer per square meter.
- We conduct experiment: we plant many plots of tomatoes, each identically tended, except that some get 100 grams of fertilizer.
- Which plot is treated with fertilizer and which is not is determined randomly. Thus, any other difference between the plots is unrelated to whether they receive fertilizer.
- At the end of the season we compare the harvest from each plot. The difference between the average yield per square meter of the treated and untreated plot is the effect of fertilizer on tomato production.

Causal Effects and Idealized Experiments

Randomized Controlled Experiments

- In this type of experiment there is a **control group** (which receives no treatment) and a **treatment group** (which is treated) and the assignment to each group is **random**.
- The random assignment eliminates the possibility of any systematic difference between the treated subjects and the control subjects, other than the treatment itself.

Causal Effects and Idealized Experiments

Randomized Controlled Experiments

- If a randomized controlled experiment is properly implemented on a large enough scale, then it will yield an estimate of the causal effect of the treatment on the outcome of interest.
- The **causal effect** is defined to be the effect on an outcome of a given action or treatment as measured in an ideal randomized controlled experiment.

Causal Effects and Idealized Experiments

Forecasting and Causality

To make a good forecast we do not need to establish a causal relationship. Multiple regression analysis allows us:

- to quantify historical relationships suggested by economic theory,
- to check whether those relationships have been stable overtime,
- to make quantitative forecasts about the future, and
- to evaluate the precision of these forecast.

Data Sources and Types

Experimental versus Observational Data

- **Experimental data** come from experiments designed to evaluate a treatment or a policy, or to investigate a causal effect.
- Real world experiments with human subjects are difficult to administer and control, and thus they are typically flawed relative to ideal randomized experiments.

Data Sources and Types

Experimental versus Observational Data

- **Observational data** is obtained from observing actual behavior outside an experimental setting by the use of surveys and administrative records.
- In observational data “treatments” are not assigned at random, and thus econometric techniques are needed to sort out the effects of the relevant variable (“treatment”) from other relevant factors.

Data Sources and Types

Data Types

- **Cross-sectional data** are data on different entities —consumers, workers, firms, etc.— for a single time period.
- **Time series data** are data for a single entity — person, firm, country, etc.—collected at multiple time periods.
- **Panel data** are data for multiple entities in which each entity is observed at two or more time periods.